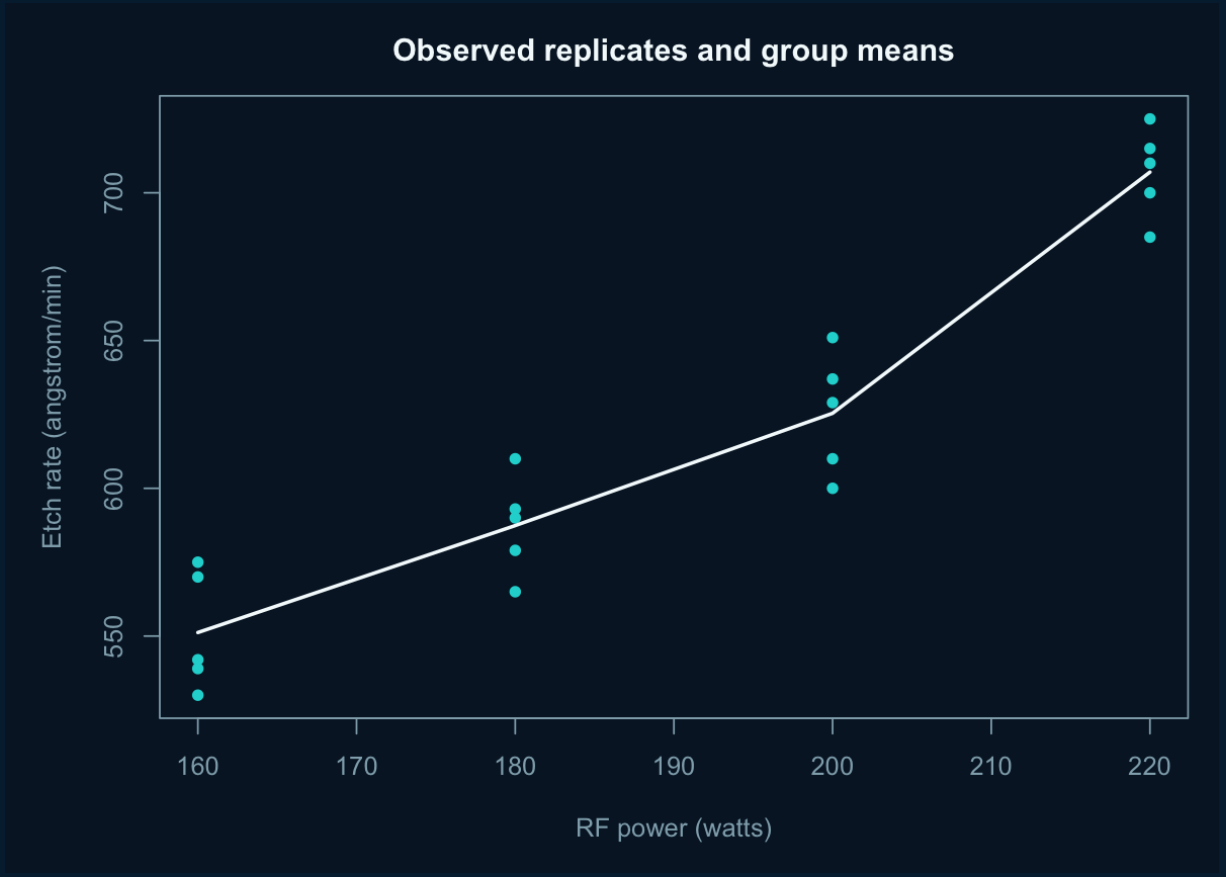
COMPLETED COURSEWORK / PORTFOLIO REVISION

Experimental Design in R

RF power and plasma etch rate

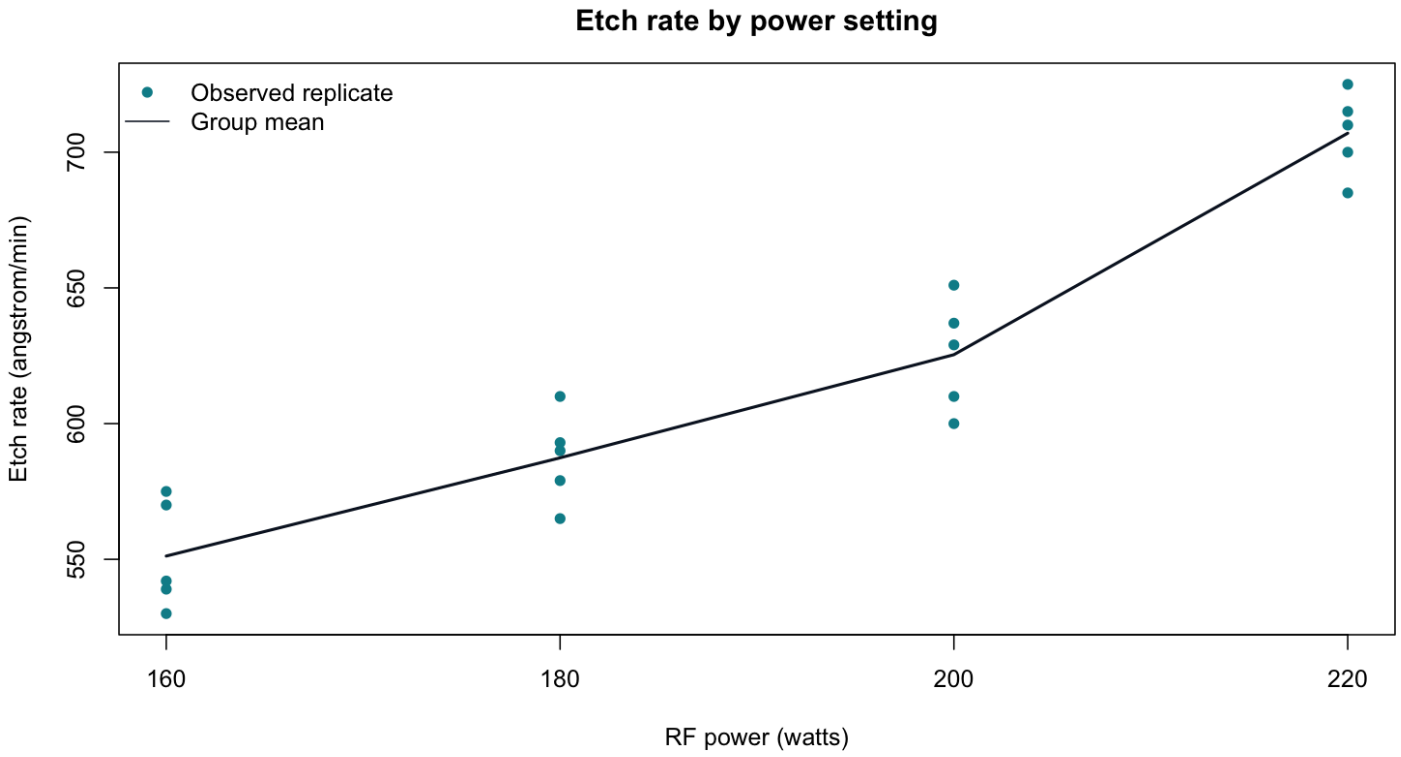
20 observations | ANOVA & Tukey comparisons



Actual analysis output | Methods and limits inside

JAMES OOKO

How does etch rate differ across RF power settings?



Mean etch rates rise from 551.2 at 160 W to 707.0 angstrom/min at 220 W. The latter is the highest tested mean, not a proven operating optimum.

Method, findings, and limits

Method

Treat power as a categorical factor. Fit a one-way ANOVA to 20 observations in four balanced groups. Inspect residuals and compare group means using Tukey-adjusted intervals.

Checked results

ANOVA: F(3,16) = 66.80, p = 2.88e-9. All six Tukey pairwise comparisons have adjusted p-values below 0.05. Shapiro-Wilk p = 0.215; Fligner-Killeen p = 0.791.

Interpretation limits

This is a textbook experiment reanalyzed for coursework, not an experiment I conducted. Independence and randomization depend on the source design. Small groups limit diagnostic power; a nonsignificant test does not prove assumptions. Results apply only to the tested settings and do not measure cost, uniformity, or wafer damage.

Source: course-provided plasma-etch numerical example from Montgomery, Design and Analysis of Experiments. Dataset values were checked against the local coursework; source documentation identifies four RF settings and five replicates.

Code and source notes: github.com/jamesooko/Final-Project-HOA732